

DTSC 5303

Cybersecurity for Data Science

FORMAT	CATEGORY	CREDITS
On-Campus	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 2 – Vital Skills (on-campus)	Machine Learning Engineer (52.5 / 100)	44.5 / 100
SPECIALIZATION General Data Science (on-campus)		

44

CURRENT AVG SCORE
across 6 job roles

82

POTENTIAL SCORE
with recommended additions

+45

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course aims to help anyone interested in data science understand the cybersecurity risks and the tools/techniques that can be used to mitigate those risks. We will cover the distinctions between confidentiality, integrity, and availability, introduce learners to relevant cybersecurity tools and techniques including cryptographic tools, software resources, and policies that will be essential to data science. We will explore key tools and techniques for authentication and access control so producers, curators, and users of data can help ensure the security and privacy of the data.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	52.5	85.2	+32.7	HIGH
AI Engineer	47.9	78.7	+30.8	HIGH
Data Scientist	43.1	87.8	+44.7	HIGH
MLOps Engineer	42.2	73.2	+31.0	HIGH
ML Engineer	41.4	82.4	+41.0	HIGH
Applied Scientist	39.9	82.2	+42.3	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

5 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 5 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

Machine Learning Engineer	52.5 → 85.2 (+32.7 pts)
web learning protection code applications systems information analysis	
AI Engineer	47.9 → 78.7 (+30.8 pts)
learning code systems based problems technology information new	
Data Scientist	43.1 → 87.8 (+44.7 pts)
toolkit analytical problems approaches	
MLOps Engineer	42.2 → 73.2 (+31.0 pts)
scientists systems learning media	
ML Engineer	41.4 → 82.4 (+41.0 pts)
media learning systems world problems new digital code	
Applied Scientist	39.9 → 82.2 (+42.3 pts)
lab computational research problems learning systems industry regulatory	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.